Xavier Rao

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Objective

Software Engineering student seeking an internship to grow technical skills and gain hands-on industry experience. Available January to August 2027.

Education

Rochester Institute of Technology (RIT)
Bachelor of Science, Software Engineering — Expected May 2028

Projects

GymTracker, *Personal Project* — August 2026 - Current

- Built a local-first iOS/Android workout tracker with custom splits, an exercise library, set logging with progression suggestions, a plate calculator, and background-accurate rest timers
- Designed an offline SQLite data layer with versioned migrations, and built analytics for volume tracking, PR detection, estimated 1RM trend charts, body weight trends, and shareable workout summary cards, plus local notifications for rest timers, weigh-ins, and inactivity
- Technologies: React Native, Expo, TypeScript, SQLite, Expo Router, EAS Build and Updates

Notescribe, *Class Project* — Spring 2026

- Designed a decoupled, event-driven backend with SQS queues and EC2 auto-scaling workers; implemented serverless APIs with API Gateway and Lambda, secured per-user file ownership with Cognito MFA, and integrated Transcribe and Bedrock to automate lecture transcription and structured study summaries
- Technologies: AWS (EC2, S3, SQS, DynamoDB, Transcribe, Bedrock, Lambda, Cognito), Terraform, Python, JavaScript

Most Likely To Game, *Personal Project* — May 2025 – April 2026

- Developed and deployed a real-time multiplayer party game with an imposter mechanic where one player secretly receives a different AI-generated question each round
- Technologies: React, Node.js, Socket.IO, JavaScript

CineVision, *Class Project* — Spring 2026

- Built and evaluated Neural Collaborative Filtering and SVD movie recommendation models against a baseline on the MovieLens 100K dataset, achieving measurable RMSE improvement
- Technologies: Python, PyTorch, scikit-learn, pandas

Work Experience

TAG Online Inc. | Clifton, NJ — Fall 2026
Intern

- Developed and maintained custom websites using WordPress, HTML, CSS, and PHP, implementing features and updates based on client requirements
- Assisted with software and systems projects, including email server management and system administration
- Researched and integrated AI tools to streamline website development workflows, improve efficiency, and automate repetitive tasks

MLC STEM Center | Bloomfield, NJ — Summer 2022/2024/2025/2026
STEM Mentor/Counselor

- Developed and delivered lesson plans on coding, physics, chemistry, and 3D design for classes of 20 students
- Designed and led week-long curriculums introducing artificial intelligence concepts through hands-on coding projects

NanoGraf | Chicago, IL — June 2021 – July 2021
Intern

- Engineered high-performance battery cells by formulating specialized slurries and applying precision compression in a controlled glove box environment

Skills

Languages: Java, Python, JavaScript, Typescript, HTML/CSS, SQL, C, PHP, Terraform
Frameworks: Angular, React.js, Node.js, Flask, Spring Boot, Express.js
Software: Visual Studio Code, IntelliJ IDEA CE, STM32CubeIDE, Git, GDB, Maven, Jacoco, AWS
Courses: Machine Learning, Intro Artificial Intelligence, Engineering Cloud Software Systems, Engineering of Software Subsystems, Software Design for Computing Systems, Web & Mobile I, Web Engineering